

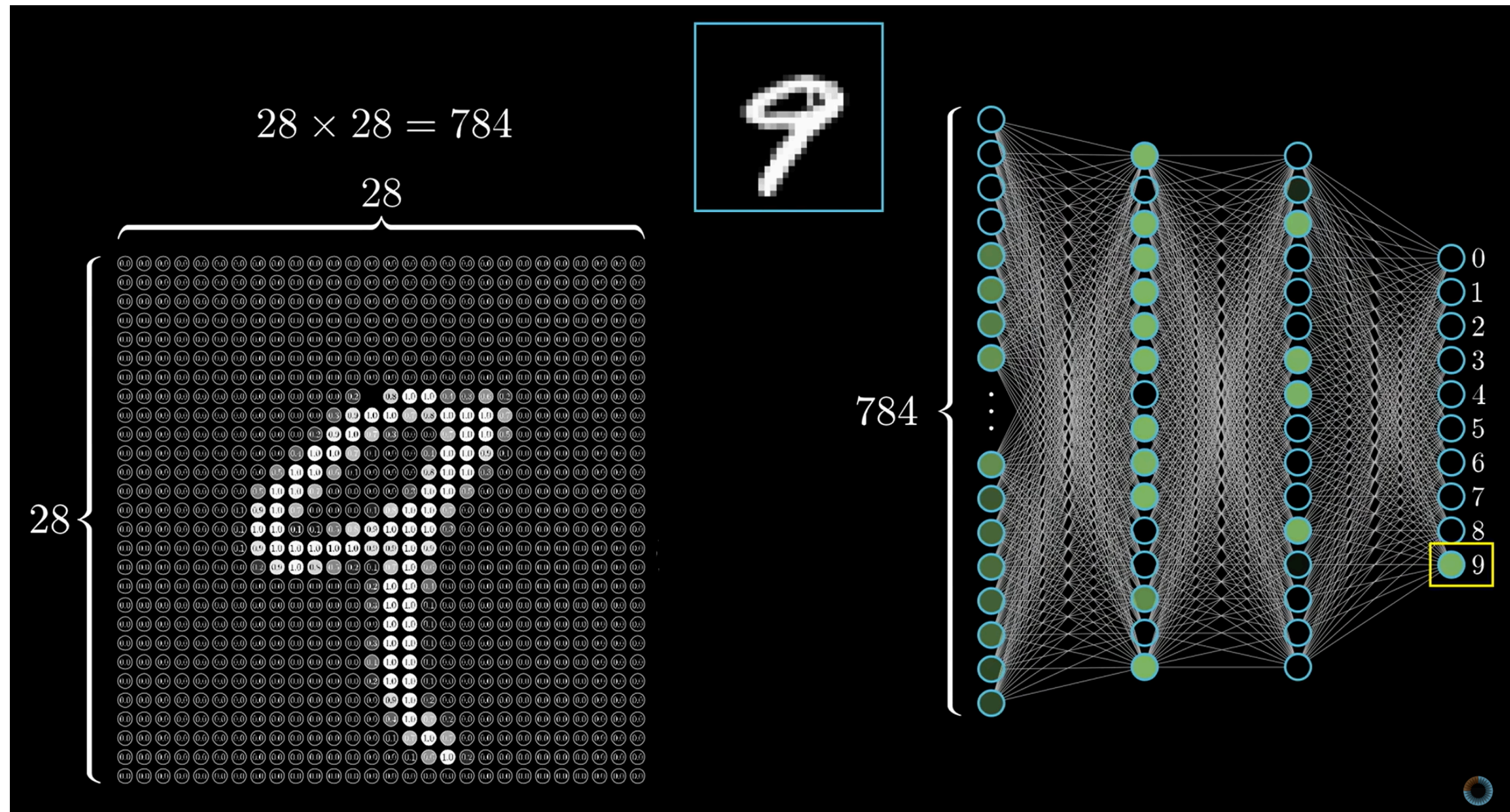
Advanced Machine Learning

DATA 642

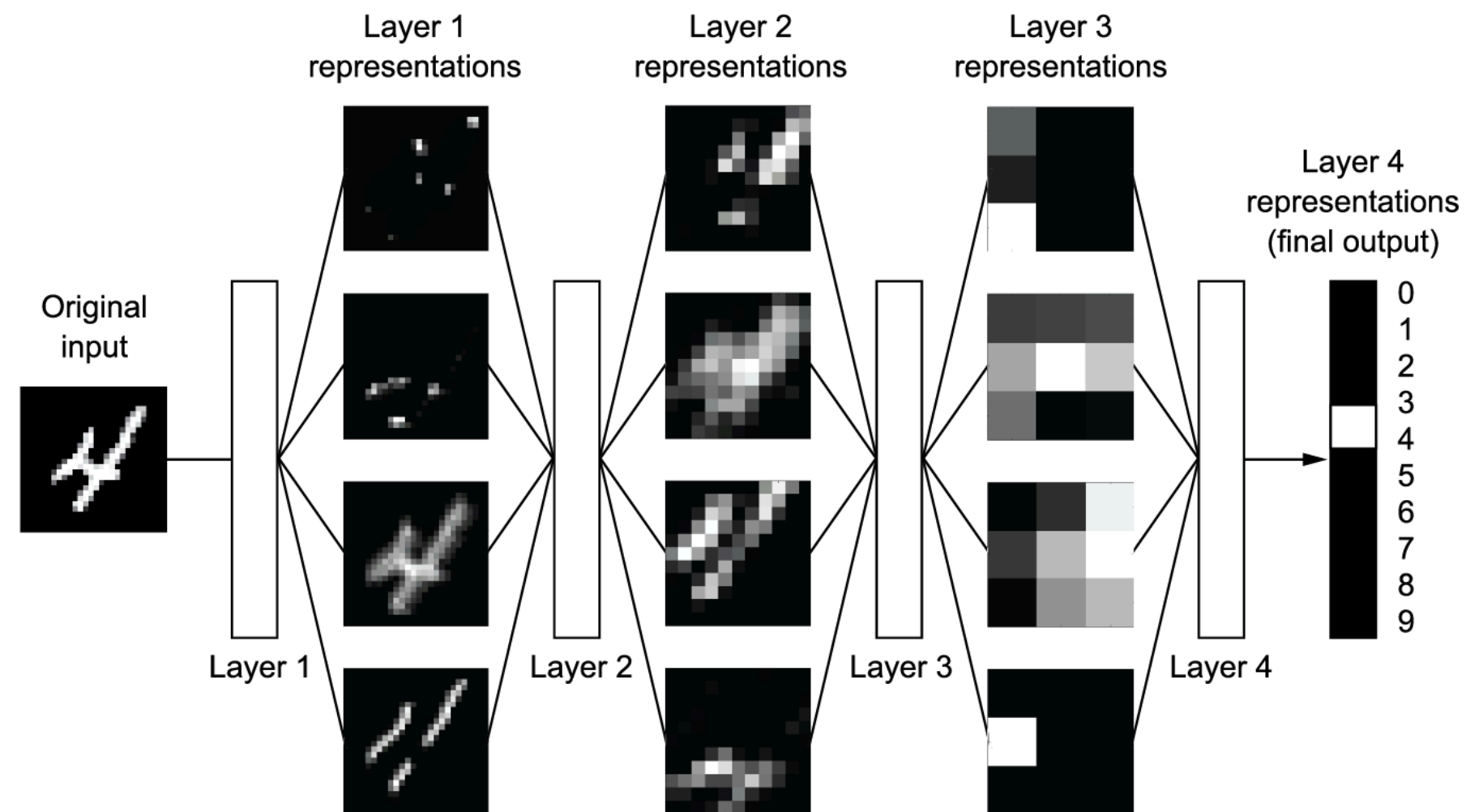
Lecture 12 — Multi layer neural networks

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier, and Hands-On Machine Learning with Scikit-Learn Keras and TensorFlow, Aurelien Geron, O'Reilly, and 3blue1brown, or 3b1b for short, which is a YouTube channel about discovery and creativity in math, with an emphasis on visualizations. <https://www.3blue1brown.com/topics/neural-networks>

Multi-Layer neural network

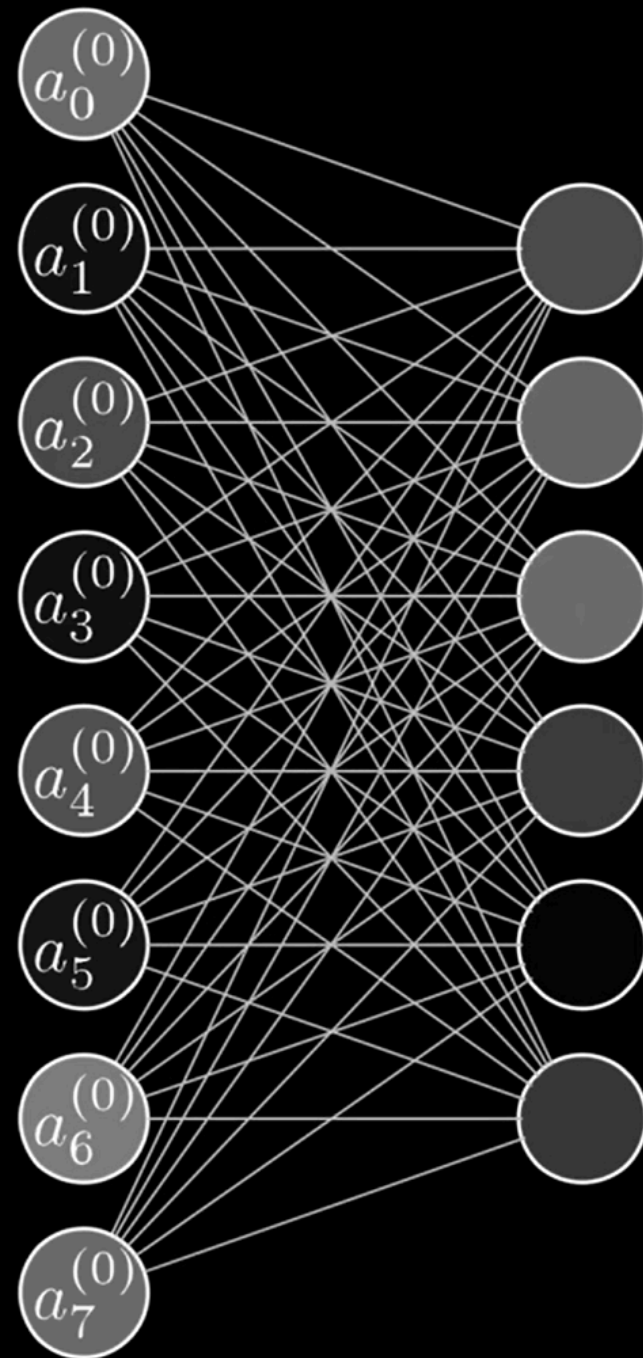


What do the layers of a deep learning algorithm look like?



Think of a deep network as a multistage information-distillation process, where information goes through successive filters and comes out increasingly purified (that is, useful with regard to some task).

Linear algebra notation in pictures



$$\mathbf{a}^{(1)} = \sigma(\mathbf{W}\mathbf{a}^{(0)} + \mathbf{b})$$

$$\sigma \left(\begin{bmatrix} w_{0,0} & w_{0,1} & \dots & w_{0,n} \\ w_{1,0} & w_{1,1} & \dots & w_{1,n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{k,0} & w_{k,1} & \dots & w_{k,n} \end{bmatrix} \begin{bmatrix} a_0^{(0)} \\ a_1^{(0)} \\ \vdots \\ a_n^{(0)} \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_n \end{bmatrix} \right)$$



Mathematics Behind layers

Linear Transformation

- Each neuron in a neural network performs a linear transformation of the input data using weights and biases.
- Mathematically, this can be represented as:

$$\mathbf{z}^{(l)} = \mathbf{W}^{(l)} \mathbf{a}^{(l-1)} + \mathbf{b}^{(l)} \quad (1)$$

where:

- $\mathbf{z}^{(l)}$ is the weighted sum of inputs and biases at layer l .
- $\mathbf{W}^{(l)}$ is the weight matrix at layer l .
- $\mathbf{a}^{(l-1)}$ is the activation output from the previous layer.
- $\mathbf{b}^{(l)}$ is the bias vector at layer l .

Output

- The output of each neuron at layer l after activation has been applied is denoted as $\mathbf{a}^{(l)}$:

$$\mathbf{a}^{(l)} = \sigma(\mathbf{z}^{(l)}) \tag{2}$$

where σ is the activation function.

Activation Function

- After the linear transformation, an activation function is applied element-wise to introduce non-linearity.
- Common activation functions include sigmoid, tanh, ReLU, and softmax.

